Adaptive RCS Architecture for Concurrent SLAM and CNN Based Autonomous Target Acquisition in GNSS-Denied Loitering Munitions

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Abstract

Modern loitering drones use Global Navigation Satellite System (GNSS) for navigation and command-control (C2) links for flight control, making them vulnerable to electronic warfare (EW). This report evaluates adaptive RCS architectures—built on FPGA-SoCs with dynamic partial reconfiguration (DPR) —to facilitate both navigation and terminal guidance when drones are jammed. Specifically this analysis includes recent advances in parallel computing of Simultaneous Location-Mapping (SLAM) and convolutional neural net based inference on shared hardware, comparison of resource-sharing strategies (spatial vs. temporal), and the measure of key metrics: latency, power draw, and targeting accuracy under simulated EW conditions. The designs in this report show that hybrid pipelines of dynamically loaded SLAM and object-detection kernels can maintain sub-meter strike precision while consuming under 10 watts. These findings demonstrate the increasing feasibility for fully autonomous, GNSS-denied loitering munitions that reconfigure their compute fabric in real time to balance performance, energy efficiency, and resilience.

Keywords— GNSS-denied navigation; reconfigurable computing systems; FPGA-SoC; dynamic partial reconfiguration; parallel SLAM and CNN; loitering munitions; autonomous terminal guidance; visual-inertial odometry; electronic warfare.

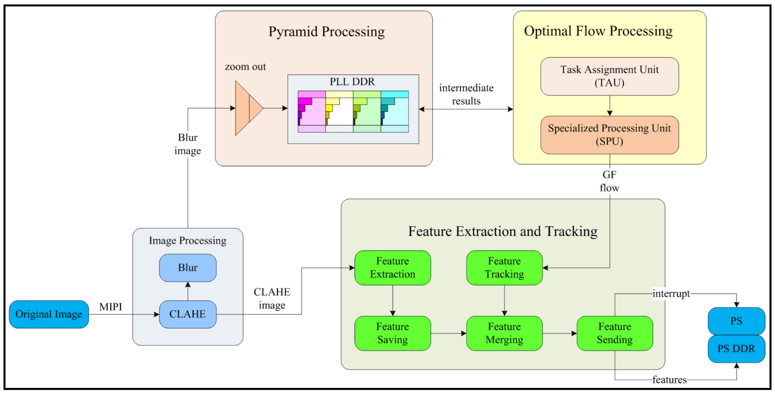
General Overview

Loitering munitions, aka “kamikaze drones” combine the endurance of UAVs with the striking capabilities of a guided missile. These drones hover over a target zone until a suitable target is found and then proceed to do a terminal dive for detonation. Under ideal circumstances, these drones would use GNSS waypoints and remote command links to navigate and receive target updates, but adversaries typically employ EW tactics including GNSS jamming, signal spoofing and C2 disruption. To avoid mission failure from jamming, drones must fuse data from onboard sensors such as barometers, magnetometers, and cameras for inertial navigation and Visual-Inertial Odometry (VIO). Simultaneously, they must perform real time target perception by running lightweight convolutional neural networks. These systems must run concurrently so the drone can compute its navigation path and terminal strike trajectory. Meeting these compute-intensive demands under strict Size, Weight, Power, and Cost (SWaP-C) constraints calls for FPGA-SoC platforms. The fine-grained parallelism and DPR allows a single chip to host multiple accelerators including one pipeline for SLAM/VIO and another for CNN inference. By swapping in only the needed logic at each mission phase, these systems maximize hardware utilization and minimize power draw.

In a GNSS-denied scenario, the drones start on GNSS-aided mode to reach the target zone. Upon entering the jamming range, the onboard FPGA reconfigures from GNSS support to the VIO kernel for navigation, while simultaneously loading a CNN accelerator for vision and target detection in 3D. This RCS approach allows for sub-meter accuracy and strike reliability, demonstrating how reconfigurable hardware and onboard AI can overcome electronic warfare threats without human intervention.

State of the Art

FPGA-SoC Architectures for Parallel SLAM and CNN



Strike drone platforms require simultaneous localization and perception under tight Size-Weight-Power-Cost constraints. FPGA-SoCs with embedded CPUs offer a promising solution of accelerating the visual SLAM pipeline (feature extraction, optical flow, pose estimation) and deep-learning inference to custom hardware blocks. For instance in the figure above, Zhang et al (2023). demonstrated millisecond-scale feature tracking on a Xilinx Zynq UltraScale+ by migrating the SLAM front-end onto FPGA logic, reducing both latency and power versus CPU-only implementations. Similarly, Névoa et al (2021). combined a hardware-accelerated PointPillars CNN for 3D object detection with CPU based LiDAR graph-SLAM on a single ZCU102 fpga, achieving real-time performance within ~10 watts. Yu et al (2020). extended this concept by deploying concurrent monocular visual-odometry and loop-closure CNNs on a Zynq MPSoC’s deep learning processing unit (DPU), using fixed-point quantization and cross-component scheduling to maintain sub-30 Hz operation for decentralized SLAM. Together, these works establish a development pattern where a dedicated fpga can be used for parallel SLAM and CNN tasks implemented either as separate pipelines or multi-core DSP clusters with the data channeled through AXI streams or shared BRAM. This enables drones to navigate and target autonomously without the need of off-board logic.

Dynamic Partial Reconfiguration for Adaptive Pipelines

As drone missions shift between loitering, navigation, and terminal guidance, a static FPGA design either over-provisions resources or under-utilizes hardware. DPR addresses this by allowing portions of the FPGA to be reprogrammed at run-time without interrupting other logic. Tang et al (2020). made a scheduling framework that partitions reconfigurable regions and chains bitstream loads via a graph-based ILP heuristic, minimizing downtime in mixed-criticality SoCs. Irmak et al. iterated on this by creating reconfigurable macroblocks for CNN accelerators. While one block executes a convolution layer, the next layer's engine is loaded at the same time. Pooling operations are folded into adjacent logic to minimize reconfiguration tasks. This pipeline design masks bitstream loading behind ongoing compute, achieving throughput on par with static designs but with the ability to quickly swap SLAM and CNN cores as mission demands change. Vendor tools such as Xilinx DFX integrate DPR scheduling into the build process, making it possible to time-share a single FPGA among high-rate visual odometry, object detection, and even RISC-V soft-core tasks, crucial for resource-constrained loitering munitions.

Low-Power, Low-Latency CNN Accelerators

Object detection CNNs like YOLO and SSD pose a challenge to small drones due to their billions of operations per inference. Recent FPGA implementations overcome this with aggressive model compression (quantization to 8–16-bit, pruning, sparsity) and custom datapaths. Li et al (2023). deployed a YOLO based object detector on a Zynq UltraScale+ using the DPU IP, achieving 30+ FPS on high-resolution aerial imagery with half the power of GPU baselines. Shi et al (2023). Optimized the energy per inference on a CNN accelerator by using a DPR on a KV260 board to time-multiplex YOLO-Tiny layers in a 5.25 W, 250 MHz design, delivering ~75 GOPS at 13.6 GOPS/W—an order-of-magnitude energy saving over static DPU approaches. Even ASIC research like Navion (2 mW VIO chip) underscores this trend of trimming image resolution, feature counts, and bit-width until the network fits under strict power constraints. Modern CNN accelerators therefore balance inference accuracy with energy-per-frame metrics, enabling >30 FPS detection on battery-powered drones, which is crucial for real-time target recognition and navigation.

GNSS-Denied Navigation: VIO, Sensor Fusion, and Autonomous Targeting

Upon GPS jamming, drones rely on multisensor fusion and blend IMU, camera, LiDAR, and radar to maintain state estimates. Visual-Inertial Odometry (VIO) fuses image features with inertial data via Extended Kalman Filters (EKF) or pose-graph optimizers. Enhancements such as ST-VIO integrate semantic segmentation to reject outliers in low-light or motion-blur circumstances. Jung et al (2019). demonstrated CNN-guided flight by detecting visual gates to steer a racing drone, foreshadowing its use to hone in on targets under GNSS denial circumstances. R-LVIO by Zhang et al. implemented LiDAR-vision-IMU fusion with cross-sensor validation and mode-switching. In feature-sparse environments, it defaults to IMU propagation or point-to-plane LiDAR matching, reducing drift by over 12% compared to prior implementations. On the hardware side, an FPGA can accelerate high-rate IMU integration or sparse point-cloud alignment to run these algorithms at hundreds of Hz. Concurrently, lightweight CNNs, optimized for 8-bit precision, scan video feeds for target signatures and map their location within the SLAM-derived point cloud to compute a terminal collision path. This merge of robust VIO, adaptive sensor fusion, and onboard AI inference—implemented on reconfigurable, low-power hardware—enables loitering drones to navigate, search, and strike autonomously, even amongst the presence of electronic warfare.

Analysis of Key Trends

To illustrate the evolution of FPGA-driven navigation and target perception for GNSS-denied drones, five works were examined, highlighting innovations in dynamic reconfiguration, CNN acceleration, sensor fusion, and hardware–algorithm co-design.

Yu et al. (2020) – CNN-based Monocular Decentralized SLAM on Embedded FPGA

Yu and colleagues integrated two CNNs—one for learning-based monocular visual odometry (VO) and another for NetVLAD-driven loop closure onto a Xilinx Zynq MPSoC's DPU. By retraining networks under 16-bit fixed-point constraints, they preserved VO accuracy while running both VO and place-recognition concurrently at over 20 FPS. Cross-component pipeline scheduling runs visual-odometry inference on the current frame at the same time it performs loop-closure on the previous frame, maximizing parallel use of the FPGA's limited resources. This approach demonstrates that fully CNN-driven SLAM—both front-end and back-end can perform in real-time on embedded hardware.

Liu et al. (2019) – eSLAM: Energy-Efficient Accelerator for ORB-SLAM

Presented at DAC 2019, eSLAM targets the ORB-SLAM front-end—FAST corner detection and BRIEF descriptor matching via hardware-aware algorithmic tweaks. By redesigning BRIEF for rotational invariance in logic and applying spatial pipelining, loop unrolling, and memory partitioning, the FPGA implementation achieved benchmark performance x3 better than the intel i7 frame rate and x31 better then ARM on the TUM RGB-D benchmark, while improving energy efficiency by x25-71. This domain-specific accelerator shows algorithm hardware co-design of SLAM on FPGA yields great savings in power which is a crucial benefit for battery-limited drones. The trade-off of modifying descriptor patterns had minimal impact on SLAM's robustness, underscoring the value of co-optimizing algorithms and hardware.

Irmak et al. (2021) – DPR for Flexible CNN Accelerators

Irmak and co-authors addressed the inflexibility of static FPGA CNN accelerators by introducing reconfigurable macroblocks and leveraging Dynamic Partial Reconfiguration (DPR). Their design assembles layer-specific blocks at runtime, then masks reconfiguration latency by overlapping bitstream loads with ongoing convolution computations. Pooling operations are fused into adjacent convolution units to reduce reconfiguration events. Evaluated on multiple LeNet variants, this DPR-enabled accelerator matched the throughput of a dual-engine static design while supporting different network structures with great flexibility. The work shows that, with careful scheduling and architecture, DPR can provide multi-model adaptability without sacrificing performance, thus enabling drones to switch between SLAM and detection tasks or between high-accuracy and low-power CNNs on the fly.

Shi et al. (2023) – EDECA: Reconfigurable YOLO-Tiny Accelerator

Targeting the compact YOLOv2-Tiny network, Shi et al (2023). combined model simplification with DPR on a Xilinx KV260 FPGA. They partitioned the network into sequential reconfigurable segments and standardized module interfaces to minimize communication overhead. By optimizing layer-execution order and streaming data to hide reconfiguration, their design achieved 75.2 GOPS at 5.25 W (13.6 GOPS/W) on a 250 MHz clock which is roughly 5× the energy efficiency of prior static DPU approaches. This "time-for-space" strategy demonstrates how DPR can allow for implementing large CNNs on small FPGAs by trading some latency for dramatic savings in power and area. This is a direct benefit for strike drones that must balance inference accuracy with extended flight endurance.

Zhang et al. (2023) – R-LVIO: Resilient LiDAR-Visual-Inertial Odometry

Zhang and colleagues advanced GNSS-denied navigation through adaptive multi-sensor fusion, combining LiDAR, camera, and IMU data in a resilient odometry pipeline. Key innovations included high-frequency IMU bias updates using rapid visual/LiDAR cues, cross-sensor outlier rejection via reprojection error checks, and a mode-switching mechanism that toggles between planar LiDAR matching and point-to-surface methods in unstructured environments. Although implemented in software, their approach suggests FPGA-based IMU integration engines and selective point-cloud processors could sustain the high update rates needed for robust drone state estimation. R-LVIO reduced drift by over 12% as validated on diverse aerial datasets. This is relative to prior VIO methods which underscored the importance of adaptable sensor-fusion filters in contested or GPS-denied settings.

Collective Trends

These five works collectively illustrate emerging themes: hybrid neural-network SLAM systems on FPGA (Yu 2020), hardware-accelerated classical SLAM (Liu 2019), dynamic reconfiguration for adaptability (Irmak 2021; Shi 2023), and advanced sensor fusion algorithms for robustness (Zhang 2023). Together, these advances suggest that next-generation strike drones will integrate custom FPGA-accelerated vision engines (both CNN-based and classical), leverage on-the-fly hardware reconfiguration to optimize performance versus resource use, and employ real-time multi-sensor fusion—offloading the heaviest computations to dedicated logic.

Summary and Comparative Analysis

Latency vs. Accuracy: Control loops demand low-latency processing—1–2 ms per frame—for stability. FPGAs excel via pipelined front-ends; heavier algorithms deliver higher accuracy. Designers run tasks (Tiny-YOLO, pruned networks) at high frame-rates and defer updates to slower threads or a CPU.

Power vs. Throughput: Energy efficiency is critical for battery UAVs. FPGA solutions deliver 10–15 GOPS/W—above CPU/GPU—but higher FPS or larger models increase power. Voltage/frequency scaling and DPR enable low-power modes or kernel reconfiguration. CNNs may run during guidance, then hardware reverts to navigation mode.

Accuracy vs. FPGA Area: On-chip resources (LUTs, DSPs, BRAM) are finite. Large CNNs or simultaneous SLAM/CNN pipelines can exceed capacity, necessitating quantization, pruning, time-multiplexing, or DPR. Parallel designs maximize throughput but require larger FPGAs (higher cost/power). Reconfigurable designs reduce area but add scheduling complexity. Algorithm–hardware co-design ensures efficient fabric use.

Resilience and Reliability: In GNSS-denied environments, multi-sensor fusion (IMU, vision, LiDAR) provides redundancy. FPGAs sustain IMU integration if vision slows or reconfigure to simpler filters when features vanish. Adaptive systems detect degenerate cases (poor lighting, textureless terrain) and reallocate resources, trading performance for reliability.

In conclusion, balancing latency, power, accuracy, and resources via DPR, custom accelerators, and sensor fusion enables FPGA platforms to deliver real-time SLAM, odometry, and target recognition concurrently. This energy-efficient approach paves the way for autonomous, GPS-independent loitering munitions capable of precise strikes under electronic warfare.

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